Simulator Computation Framework

Readable sample calculation sheet for the Haul Road Rolling Resistance model

This version is organized for print and screen reading. Method B now follows the paper directly using Eq. (1), Eq. (3), Eq. (5), Eq. (6), and Eq. (8) through Eq. (11), while Method A remains the fast operational heuristic.

Tire Load: 900 kN | Tire Stiffness: 13 kN/mm | Ground Stiffness: 4.5 kPa/mm | Derived Width: 0.35 m

Initial State Parameters

TIRE LOAD

900 kN

Dynamic load transferred to an individual tire from the suspension strut.

TIRE STIFFNESS

13 kN/mm

Effective tire stiffness corresponding to the mapped 7% diametral strain condition.

GROUND STIFFNESS

4.5 kPa/mm

Resilient pressure stiffness derived from the cyclic plate load test and used directly in Eq. (6).

METHOD A

Equivalent Linear Interaction Model

KEY EQUATION

Method A rolling resistance

$$RR\% = 2.0 + (\delta_{ground} \times 0.2) + (\delta_{tire} \times 0.05)$$

01 Tire deflection

Elastic tire compression under the applied wheel load.

$$\delta_{tire} = \frac{F_i}{k_{tire}} = \frac{900}{13} = 69.23 \text{ mm}$$

Compression = 69.23 mm

02 Equivalent ground deflection

A proportional field-friendly map representing the ground response.

$$\delta_{ground} = (\frac{F_i}{k_p}) \times 0.15 = (\frac{900}{4.5}) \times 0.15 = 30.0 \text{ mm}$$

Ground sink = 30.0 mm

03 Linear rolling resistance

Baseline hard-surface resistance plus tire and ground deformation penalties.

$$RR\% = 2.0 + (30.0 \times 0.2) + (69.23 \times 0.05) = 11.46\%$$

RR_linear = 11.46%

METHOD B

Dynamic Boussinesq Solution

KEY EQUATION

Paper Eq. (1)

$$RR\% = 100 \left(\frac{y}{(\frac{\phi_{\text{tire}}}{2} - \delta_{\text{tire}})} \right)$$

01 Footprint geometry

Eq. (5) gives the area and A = Lw provides the base contact length used

$$A = 1.35\, \delta_{tire} \phi = 1.35 \times 0.069 \times 3.84 = 0.359$$

$$L = 2\sqrt{\phi\, \delta_{tire} - \delta_{tire}^2} = 2\sqrt{(3.84)(0.069) - (0.069)^2} =$$

A = 0.359 m², L = 1.022 m

02 Effective width and ground deformation

Width follows from A = Lw, then Eq. (6) uses the footprint area and res stiffness k_p.

$$w = \frac{A}{L} = \frac{0.359}{1.022} = 0.351 \text{ m}$$

$$\delta_{ground} = \frac{F_i}{A\, k_p} = \frac{900}{0.359 \times (4.50 \times 1000)} = 0.55$$

w = 0.351 m, δ_ground = 557.3 mm

03 Lead contact geometry

Eq. (8), Eq. (9), Eq. (10), and Eq. (11) define the geometry of the leadin contact.

$$\beta = 30.87^\circ,\ \alpha = 31.38^\circ,\ \omega = 32.45^\circ$$

$$l_{arc} = 1.039 \text{ m}, \quad l_{arc} \cos \omega = 0.876 \text{ m}$$

l_arc = 1.039 m

04 Reaction offset and rolling resistance

Eq. (3) gives y and Eq. (1) converts that offset into equivalent slope perc

$$y = 0.492 \text{ m}$$

$$RR\% = 100\left(\frac{0.492}{\left(\frac{3.84}{2} - 0.069\right)}\right) = 26.58\%$$

RR_paper = 26.58%

Prepared for web download and print review. All equations are rendered in KaTeX for consistent browser and PDF readability.